

Numerical Modelling using Monte Carlo method

Results: -

Using the simulation at $T=600\text{K}$, the second virial coefficients for the gases are: -

- Helium: $-1.228889\text{e-}05\text{ m}^3/\text{mol}$
- Nitrogen: $2.233479\text{e-}05\text{ m}^3/\text{mol}$
- Argon: $-1.213656\text{e-}05\text{ m}^3/\text{mol}$
- Xenon: $-1.734011\text{e-}05\text{ m}^3/\text{mol}$

Ideal Gas Molar Volume of He at 600K : $8.314000\text{e-}05\text{ m}^3/\text{mol}$

Virial Gas Molar Volume of He at 600K : $9.365834\text{e-}05\text{ m}^3/\text{mol}$

Summary of Results

-Calculated Correction Scores (B_2): The algorithm successfully generated B_2 values for Helium, Nitrogen, Argon, and Xenon. It proved that larger atoms (like Xenon) have a stronger attractive pull, while tiny atoms (like Helium) are dominated by repulsive forces.

-Predicted Real Molar Volume: At extreme pressures (600 bar), the Ideal Gas Law predicts a small volume. By applying our calculated B_2 score, the simulation proved the real volume is significantly larger, because the compressed atoms violently repel each other and force the gas to expand.

-Algorithm Validation: The simulation proved that scaling up the number of random iterations directly reduces variance and aligns the computational output with standard laboratory literature.